

Compiler Construction - Final - Spring 2025

PART 1: (given CFG)

$MST \rightarrow S \mid a \mid List \mid Mp$

$List \rightarrow id \mid id, List$

$M \rightarrow M$

$T \rightarrow (S) \mid$

$S \rightarrow b, S \mid b$

Question 1:

a) Apply Top-Down Parsing to Find First() & Follow() of all non-terminals

b) Apply Top-Down Parsing to Construct LL(1) Predictive Parsing Table.

Question 2:

Apply Top-Down Parsing to perform Stack processing of any supported string by given rule.

PART 2: (given CFG)

$F \rightarrow \text{for}(D; C; I) \{ B \}$

$B \rightarrow x$

$D \rightarrow TL;$

$T \rightarrow \text{int} \mid \text{char}$

$L \rightarrow \text{id} \mid \text{id}, L$

$C \rightarrow 0 \mid 1$

$I \rightarrow \text{id}++$

Question 3:

Apply Bottom-Up Parsing to Construct SLR Parser to Construct Deterministic Flow and SLR Parsing Table.

Question 4:

Apply Bottom-up Parsing to perform SLR Stack Processing of any complex string of below flow